

A
BRIGHTER
YOU
THROUGH
KNOWLEDGE

LEARN
PLAN
PRACTICE
GROW

Data Science & Machine Learning Roadmap Planner

*Same
Curiosity
Bigger
Possibilities*

A structured learning guide with roadmap, chapter breakdowns, milestones, and notes

SKILLS
IDEAS
PROGRESS
A BRIGHTER
TOMORROW



Designed & Curated by Atefe Asadi

THE DATA SCIENCE & MACHINE LEARNING ROADMAP PLANNER

A 24-week, project-based learning system aligned with recurring topics in Stanford, Harvard, Coursera programs, and major data-science books.

DESIGN

Minimal editorial workbook
24 weeks · 14 chapters · capstone

CREATED BY

Atefe Asadi

CURRICULUM VALIDATION

What “verified” means here

This planner is independently designed. It is not approved, certified, sponsored, or endorsed by Stanford, Harvard, Coursera, Google, O’Reilly, Wiley, or Udemy.

The core sequence was checked against official/public course syllabi and publisher tables of contents. Where a topic is not clearly part of those core sources, it is labeled as an optional extension rather than presented as university-validated content.

AREA	PRIMARY EVIDENCE	SECONDARY SUPPORT	STATUS
Python	Harvard CS50P	Python for Data Analysis; Data Science from Scratch	Verified
Reproducibility	Harvard Data Science certificate	CS50 / GitHub workflow	Verified
Probability	Stanford CS109; Harvard Stat 110	Practical Statistics	Verified
Inference + regression	Harvard Data Science; Google Advanced Analytics	Practical Statistics	Verified
Machine learning	Stanford CS229; Coursera ML Specialization	Data Science from Scratch	Verified
Visualization + communication	Harvard Data Science	Storytelling with Data	Verified
Capstone	Harvard Data Science Capstone; Google Capstone	Data Science for Business	Verified
Time series / NLP / XAI	Not core to the validated backbone	Use as elective extensions	Optional

Udemy note: Udemy courses vary by instructor and are useful as supplementary practice, but they are not used here as evidence that a topic is academically “approved.”

MASTER ROADMAP

Learn in this order

The sequence follows prerequisite logic: programming and data → mathematical/statistical foundations → modeling → validation → communication and capstone.

24-WEEK LEARNING ROADMAP

Build foundations → work with data → learn statistics → model → validate → communicate.



EVIDENCE

Every phase ends with a public or reviewable artifact.

RULE

Do not move forward only because you finished videos.

REVIEW

Every 4 weeks: revise portfolio, notes, and weak concepts.

Rule: if you can demonstrate a prerequisite with a small diagnostic project, compress that chapter and spend the saved time on modeling, research, or portfolio work.

24-WEEK PLAN

Six months, one visible output every week

WK	FOCUS	CORE WORK	EVIDENCE
1	Python core	Functions + control flow	Mini script
2	Python data structures	Files + tests + Jupyter	Mini project
3	Git + reproducibility	CLI + env + GitHub	Reproducible repo
4	SQL core	Filtering + aggregation	Query set
5	SQL joins + windows	Data quality checks	SQL case study
6	NumPy + pandas	DataFrame fluency	Wrangling notebook
7	Cleaning + reshape	Missing data + dates	Clean dataset
8	EDA + storytelling	Purposeful charts	EDA report
9	Math for ML	Vectors + gradients	Math notebook
10	Probability I	Conditioning + Bayes	Simulation exercises
11	Probability II	Distributions + CLT	Probability notebook
12	Inference	CIs + tests	Inference memo
13	Experiments	Power + A/B testing	Experiment plan
14	Linear regression	Diagnostics + interpretation	Regression report
15	Logistic + regularization	Classification baseline	Model report
16	ML workflow	Splits + CV + metrics	Reusable ML template
17	Trees + forests	Model comparison	Tree benchmark
18	Boosting + SVM	Tuning + calibration	Supervised study
19	Clustering	Scaling + k-means	Segmentation analysis
20	EM + PCA + anomaly	Representation	Unsupervised report
21	Neural networks	Baseline vs MLP	NN benchmark
22	Capstone design	Question + protocol	One-page protocol
23	Capstone build	Analysis + validation	Repository + draft
24	Capstone publish	Story + limitations	Report + presentation

01

Python & Computational Thinking

WEEKS 1-2

STEP	SUBCHAPTER	LEARN
1.1	Core Python	Variables, types, expressions, functions, scope
1.2	Control flow	Conditionals, loops, iteration patterns
1.3	Data structures	Lists, tuples, dictionaries, sets, comprehensions
1.4	Errors & testing	Exceptions, debugging, assertions, unit-test basics
1.5	Files & modules	File I/O, imports, packages, reusable scripts
1.6	Jupyter workflow	Notebooks, cells, reproducible narrative analysis
SOURCE ALIGNMENT		QUICK NOTES
Harvard: CS50P — Introduction to Programming with Python		
Book: Data Science from Scratch		
Book: Python for Data Analysis		
EVIDENCE / DELIVERABLE		
A small Python repository with reusable functions, tests, a README, and one notebook.		
MENTOR NOTE <i>If Python is already comfortable, use this as a diagnostic week rather than a long review.</i>		
		☐ learned ☐ practiced
		☐ documented ☐ published

02

Developer Workflow & Reproducibility

WEEK 3

Good data science is not just analysis; it must be reproducible, reviewable, and easy to rerun.

[illegible]

03

SQL & Relational Data

WEEKS 4-5

Most real analytical work starts with extracting trustworthy data, not with fitting a model.

[illegible]

04

NumPy, pandas & Data Wrangling

WEEKS 6-7

Transform raw data into analysis-ready data without losing meaning or traceability.

STEP	SUBCHAPTER	LEARN
4.1	NumPy arrays	Shapes, dtypes, vectorization, broadcasting, indexing
4.2	DataFrames	Selection, filtering, assignment, sorting
4.3	Group operations	groupby, agg, transform, pivot
4.4	Missing data	Detection, missingness patterns, justified handling
4.5	Merge & reshape	merge, concat, wide/long reshaping
4.6	Dates & strings	Datetime parsing, text cleaning, categorical types
SOURCE ALIGNMENT Book: Python for Data Analysis — Wes McKinney Harvard: Data Science — wrangling concepts EVIDENCE / DELIVERABLEA cleaning notebook that exports a clean dataset, data dictionary, and transformation log. MENTOR NOTE<i>Never hide a transformation. Every important cleaning choice should be explainable.</i>		QUICK NOTES ☐ learned ☐ practiced ☐ documented ☐ published

05

EDA, Visualization & Data Storytelling

WEEK 8

Before modeling, understand distributions, anomalies, relationships, and what the audience actually needs to see.

STEP		SUBCHAPTER	LEARN
5.1		Descriptive statistics	Center, spread, quantiles, robust summaries
5.2		Distributions	Histograms, density, boxplots, skew and tails
5.3		Relationships	Correlation, scatterplots, grouped comparisons
5.4		Chart selection	Choose visual form based on the analytical question
5.5		Declutter & focus	Reduce noise, guide attention, use color intentionally
5.6		Narrative	Context → evidence → implication → recommendation
SOURCE ALIGNMENT Harvard: Data Science — Visualization Book: Storytelling with Data Book: Practical Statistics for Data Scientists EVIDENCE / DELIVERABLE An EDA report with 5–7 purposeful charts plus a concise written narrative. MENTOR NOTE <i>A chart is not finished until a reader can tell why it matters.</i>			QUICK NOTES <

06

Math Essentials for Machine Learning

WEEK 9

You do not need pure-math depth, but you do need enough linear algebra and calculus to understand models and optimization.

[illegible]

07

Probability

WEEKS 10-11

Probability is the language behind uncertainty, statistical inference, and many machine-learning models.

[illegible]

08

Statistical Inference & Experimentation

WEEKS 12-13

Inference lets you separate signal from sampling variation and make defensible claims from limited data.

[illegible]

WEEKS 14-15

Regression connects statistical explanation with prediction and remains a critical baseline for ML.

STEP		SUBCHAPTER	LEARN
9.1	Simple regression		Slope, intercept, residuals, R^2
9.2	Multiple regression		Adjustment, confounding, interactions
9.3	Diagnostics		Residual behavior, outliers, influential observations
9.4	Regularization		Ridge, lasso, bias–variance intuition
9.5	Logistic regression		Log-odds, classification probabilities, thresholds
9.6	Interpretation		Coefficients, uncertainty, limitations, prediction vs explanation
SOURCE ALIGNMENT Stanford: CS229 — linear and logistic regression Coursera / Google: Advanced Data Analytics — regression analysis Book: Practical Statistics for Data Scientists EVIDENCE / DELIVERABLE A regression report with diagnostics, interpretation, predictive evaluation, and limitations. MENTOR NOTE <i>Use regression as a transparent baseline before reaching for a more complex model.</i>			QUICK NOTES ☐ learned ☐ practiced ☐ documented ☐ published

10

Machine Learning Workflow & Evaluation

WEEK 16

Model quality depends as much on framing, splitting, validation, and metrics as on the algorithm itself.

[illegible]

11

Supervised Machine Learning

WEEKS 17-18

Compare model families using the same validation design instead of choosing algorithms by popularity.

[illegible]

12

Unsupervised Learning & Representation

WEEKS 19-20

When labels do not exist, structure can still be discovered—but interpretation and stability matter.

[illegible]

13

Neural Networks & Modern ML Foundations

WEEK 21

Understand the core mechanics of neural networks, then decide whether they are justified for the problem and data size.

STEP	SUBCHAPTER	LEARN
13.1	Network structure	Layers, weights, biases, activations
13.2	Loss & optimization	Loss functions, gradient descent, backpropagation
13.3	Regularization	Early stopping, dropout intuition, weight penalties
13.4	Model diagnostics	Training curves, under/overfitting, baselines
13.5	Responsible ML	Bias, subgroup behavior, data provenance, misuse risk
13.6	Optional directions	Recommenders and reinforcement-learning overview
SOURCE ALIGNMENT Stanford: CS229 — neural networks Coursera: Machine Learning Specialization — neural nets, recommenders, RL EVIDENCE / DELIVERABLE A small neural network benchmarked against a simpler baseline, with a short complexity justification. MENTOR NOTE <i>Deep learning is not the default answer for small tabular datasets.</i>		QUICK NOTES ☐ learned ☐ practiced ☐ documented ☐ published

14

Communication, Business Thinking & Capstone

WEEKS 22-24

A strong data scientist must connect a valid analysis to a decision, research question, or usable product.

STEP	SUBCHAPTER	LEARN
14.1	Problem statement	Research/business question, population, outcome, decision
14.2	Data provenance	Source, inclusion criteria, data quality, ethics
14.3	Analysis protocol	Baseline, methods, validation, assumptions
14.4	Communication	Audience, chart choice, narrative, recommendation
14.5	Reproducibility	Environment, seeds, README, rerun instructions
14.6	Portfolio delivery	Repository, report, presentation, limitations, next steps
SOURCE ALIGNMENT		QUICK NOTES _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____
Harvard: Data Science: Capstone		
Coursera / Google: Advanced Data Analytics Capstone		
Book: Data Science for Business		
Book: Storytelling with Data		
EVIDENCE / DELIVERABLE		
A portfolio-grade capstone repository + 2–4 page report + 5-minute presentation.		
MENTOR NOTE <i>A capstone should show judgment, not just code volume.</i>		
		☐ learned ☐ practiced ☐ documented ☐ published

ELECTIVE EXTENSIONS

Add these after the core, not before it

These topics are useful, but they are not treated as part of the fully validated backbone above. Choose them based on your research or job target.

TRACK	LEARN	WHEN IT EARNS A PLACE
Time Series	Temporal dependence, lags, rolling features, forecasting validation	Useful for forecasting, operations, finance, longitudinal signals.
NLP	Text cleaning, vectorization, embeddings, classification, transformers overview	Useful for support tickets, research text, documents, social data.
Explainable AI	Permutation importance, SHAP concepts, partial dependence, calibration	Useful when decisions need transparency and auditability.
Biomedical Data Science	Clinical prediction, missingness, calibration, subgroup analysis, longitudinal data	Best fit for health and biostatistics research.

QUICK NOTES

READING SHELF

Five books, placed where they help most

The five-book set from your original material is retained, but mapped to the roadmap by function rather than treated as five books to read cover-to-cover at once.

#	BOOK	AUTHOR(S)	USE	FOCUS
01	Python for Data Analysis	Wes McKinney	Ch. 4	pandas, NumPy, data cleaning, DataFrame workflows
02	Storytelling with Data	Cole Nussbaumer Knafl	Ch. 5 + 14	chart choice, clutter reduction, attention, narrative
03	Practical Statistics for Data Scientists	Peter Bruce, Andrew Bruce, Peter Gedeck	Ch. 5 + 8 + 9	EDA, sampling, experiments, regression, applied statistics
04	Data Science for Business	Foster Provost, Tom Fawcett	Ch. 14	data-analytic thinking, model value, decisions, business framing
05	Data Science from Scratch	Joel Grus	Ch. 1 + 6 + 10–13	Python foundations, math, algorithms, ML intuition, neural nets

Reading strategy: use books as references attached to a project or chapter. Do not pause the roadmap to finish an entire book before building anything.

SOURCES & PROVENANCE

Official alignment sources

Checked against official/public pages in September 2026. Course contents can change, so these links are references, not permanent guarantees.

- [Stanford CS109 — Probability for Computer Scientists](#)
- [Stanford CS229 — Machine Learning](#)
- [Harvard CS50P — Introduction to Programming with Python](#)
- [Harvard Statistics 110 — Probability](#)
- [Harvard Professional Certificate in Data Science](#)
- [Coursera / Stanford Online + DeepLearning.AI — Machine Learning Specialization](#)
- [Coursera / Google — Advanced Data Analytics Professional Certificate](#)
- [O'Reilly — Python for Data Analysis, 3rd ed.](#)
- [O'Reilly — Practical Statistics for Data Scientists, 2nd ed.](#)
- [O'Reilly — Data Science from Scratch, 2nd ed.](#)
- [O'Reilly — Data Science for Business](#)
- [Wiley — Storytelling with Data](#)

FINAL REFLECTION

What changed because I completed this roadmap?

Which concepts can I now explain without notes?

Which projects best prove my skill rather than just list it?

Where do I still rely on tutorials instead of independent reasoning?

Which statistical or ML assumptions do I check automatically now?

What is the next specialization that fits my research or career direction?

Designed & curated by Atefe Asadi

Independent learning planner · curriculum-aligned · project-based